Conic Sections General: Questions and Full Solutions

Question 01

English

Let C be an ellipse having a focus at the origin O and the corresponding directrix parallel to the y-axis. If C passes through the points $A(4,3)$ and $B(0,-4)$, then the x-coordinate of the vertex of C nearer to this directrix is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C एक दीर्घवृत्त है जिसकी एक नाभि मूलबिन्दु O पर है तथा संगत नियता y-अक्ष के समांतर है। यदि C बिन्दुओं $A(4,3)$ तथा $B(0,-4)$ से होकर जाता है, तो C के उस शीर्ष का x-निर्देशांक, जो इस नियता के अधिक निकट है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since the directrix is parallel to the y-axis, let it be $x = d$ with $d \neq 0$ (the focus is not on the directrix), and let e be the eccentricity, $0 < e < 1$ for an ellipse.

For every point P of C, $PS = e \cdot PM$, where $S = O$ is the focus and PM is the distance of P from $x = d$.

At $A(4,3)$: $OA = 5$ and the distance from the directrix is $|4 - d|$, so $5 = e|4 - d|$.

At $B(0,-4)$: $OB = 4$ and the distance is $|d|$, so $4 = e|d|$.

Dividing, $\dfrac{5}{4} = \dfrac{|4-d|}{|d|}$, i.e. $5|d| = 4|4-d|$, so $5d = 4(4-d)$ or $5d = -4(4-d)$.

First case: $9d = 16$, $d = \dfrac{16}{9}$, $e = \dfrac{4}{|d|} = \dfrac{9}{4} > 1$. This conic is a hyperbola, so it is rejected.

Second case: $5d = -16 + 4d$, $d = -16$, $e = \dfrac{4}{16} = \dfrac{1}{4} < 1$, an ellipse. Check at A: $\dfrac{1}{4} \cdot 20 = 5$.

Equation: $x^2 + y^2 = \dfrac{1}{16}(x+16)^2$, i.e. $16x^2 + 16y^2 = x^2 + 32x + 256$, i.e. $15x^2 - 32x + 16y^2 = 256$.

The major axis passes through the focus perpendicular to the directrix, so it is the x-axis. Putting $y = 0$: $15x^2 - 32x - 256 = 0$, $x = \dfrac{32 \pm \sqrt{1024 + 15360}}{30} = \dfrac{32 \pm 128}{30}$, so $x = \dfrac{16}{3}$ or $x = -\dfrac{16}{5}$.

The directrix is $x = -16$, so the nearer vertex is $\left(-\dfrac{16}{5}, 0\right)$, whose x-coordinate is -3.2.

Final Answer: -3.20

Question 02

English

For integers p and q with $1 \leq p \leq 20$ and $1 \leq q \leq 20$, let

$$E_{p,q} = \{z \in \mathbb{C} : |z-p| + |z-qi| = 20\}, \qquad H_{p,q} = \{z \in \mathbb{C} : \big||z-p| - |z-qi|\big| = 10\},$$

where $i = \sqrt{-1}$. Let N be the number of ordered pairs (p,q) for which $E_{p,q}$ is an ellipse and $H_{p,q}$ is a hyperbola. Then the value of N is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

पूर्णांकों p तथा q के लिए, जहाँ $1 \leq p \leq 20$ तथा $1 \leq q \leq 20$ हैं, माना कि

$$E_{p,q} = \{z \in \mathbb{C} : |z-p| + |z-qi| = 20\}, \qquad H_{p,q} = \{z \in \mathbb{C} : \big||z-p| - |z-qi|\big| = 10\},$$

जहाँ $i = \sqrt{-1}$ है। माना कि N उन क्रमित युग्मों (p, q) की संख्या है जिनके लिए $E_{p,q}$ एक दीर्घवृत्त है तथा $H_{p,q}$ एक अतिपरवलय है। तब N का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $E_{p,q}$ and $H_{p,q}$ the fixed points are $z_1 = p$ and $z_2 = qi$, and $|z_1 - z_2| = \sqrt{p^2+q^2}$, which is $2c$ for both loci.

$E_{p,q}$: $|z-z_1| + |z-z_2| = 2a$ with $2a = 20$. By the triangle inequality $|z-z_1| + |z-z_2| \geq |z_1 - z_2|$. If $\sqrt{p^2+q^2} < 20$ the locus is an ellipse with eccentricity $e = \dfrac{\sqrt{p^2+q^2}}{20} < 1$; if $\sqrt{p^2+q^2} = 20$ the locus is the segment $z_1 z_2$ (a degenerate conic, $e = 1$), and if $\sqrt{p^2+q^2} > 20$ the locus is empty.

$H_{p,q}$: $\big||z-z_1| - |z-z_2|\big| = 2a$ with $2a = 10$. Since $\big||z-z_1| - |z-z_2|\big| \leq |z_1 - z_2|$: if $\sqrt{p^2+q^2} > 10$ the locus is a hyperbola with $e = \dfrac{\sqrt{p^2+q^2}}{10} > 1$; if $\sqrt{p^2+q^2} = 10$ it is the pair of rays on the line $z_1 z_2$ outside the segment (degenerate); if smaller it is empty.

Hence both conditions hold exactly when $100 < p^2 + q^2 < 400$.

Count of pairs with $p^2 + q^2 < 400$: for each p the number of $q \in \{1, \ldots, 20\}$ with $q^2 < 400 - p^2$ is, for $p = 1, \ldots, 19$: $19, 19, 19, 19, 19, 19, 18, 18, 17, 17, 16, 15, 15, 14, 13, 11, 10, 8, 6$ (none for $p = 20$). The total is 292; the pairs $(12, 16)$ and $(16, 12)$ with $p^2 + q^2 = 400$ are not included.

Count of pairs with $p^2 + q^2 \leq 100$: for $p = 1, \ldots, 9$: $9, 9, 9, 9, 8, 8, 7, 6, 4$ (none for $p \geq 10$), total 69. This includes $(6, 8)$ and $(8, 6)$, for which $H_{p,q}$ is degenerate.

Therefore $N = 292 - 69 = 223$.

Final Answer: 223

Question 03

English

Let E be the ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ and, for a non-zero real number k, let H_k be the hyperbola $xy = k$. Let S be the set of all non-zero real numbers k for which E and H_k have exactly two common tangents. Then the product of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E दीर्घवृत्त $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ है तथा किसी शून्येतर वास्तविक संख्या k के लिए H_k अतिपरवलय $xy = k$ है। माना कि S उन सभी शून्येतर वास्तविक संख्याओं k का समुच्चय है जिनके लिए E तथा H_k की ठीक दो उभयनिष्ठ स्पर्श रेखाएँ हैं। तब S के सभी अवयवों का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A vertical line $x = t$ meets $xy = k$ in exactly one point without touching it, and a horizontal line likewise, so neither is tangent to H_k. Hence every common tangent has the form $y = mx + c$ with $m \neq 0$.

Condition for tangency to E (with $a^2 = 9$, $b^2 = 4$): $c^2 = a^2m^2 + b^2 = 9m^2 + 4$.

Condition for tangency to H_k: substituting $y = mx + c$ in $xy = k$ gives $mx^2 + cx - k = 0$, which must have equal roots: $c^2 + 4mk = 0$, i.e. $c^2 = -4mk$.

Eliminating c^2: $9m^2 + 4km + 4 = 0$. A root $m = 0$ is impossible (the constant term is 4), and for each real root m the value $c^2 = 9m^2 + 4 > 0$ gives two distinct lines $c = \pm\sqrt{9m^2 + 4}$. (For such m, $-4mk = 9m^2 + 4 > 0$ automatically.)

The discriminant of the quadratic in m is $16k^2 - 144$.

If $|k| > 3$: two distinct real slopes, hence four common tangents. If $|k| < 3$: no real slope, no common tangent (then H_k cuts E). If $|k| = 3$: one double root $m = -\dfrac{4k}{18} = -\dfrac{2k}{9}$, giving exactly two common tangents (for $k = 3$: $y = -\dfrac{2}{3}x \pm 2\sqrt{2}$).

So $S = \{3, -3\}$ and the product of its elements is -9.

Final Answer: -9

Question 04

English

Let P be the parabola $y^2 = 4x$ and C the circle $x^2 + y^2 - 2x - 4 = 0$. The common tangents of P and C meet at the point T, and they touch C at the points B_1 and B_2. If K is the centre of C, then the area of the quadrilateral TB_1KB_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P परवलय $y^2 = 4x$ है तथा C वृत्त $x^2 + y^2 - 2x - 4 = 0$ है। P तथा C की उभयनिष्ठ स्पर्श रेखाएँ बिन्दु T पर मिलती हैं तथा C को बिन्दुओं B_1 तथा B_2 पर स्पर्श करती हैं। यदि K, C का केन्द्र है, तो चतुर्भुज TB_1KB_2 का क्षेत्रफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Completing squares, $C : (x - 1)^2 + y^2 = 5$, so $K = (1, 0)$ and the radius is $r = \sqrt{5}$. Note that K is the focus of $y^2 = 4x$.

Every tangent of P other than $x = 0$ is $y = mx + \dfrac{1}{m}$, $m \neq 0$. The line $x = 0$ is at distance $1 \neq \sqrt{5}$ from K, so it is not tangent to C.

The line $mx - y + \dfrac{1}{m} = 0$ touches C when its distance from K equals $\sqrt{5}$: $\dfrac{|m + \frac{1}{m}|}{\sqrt{m^2 + 1}} = \dfrac{m^2 + 1}{|m|\sqrt{m^2 + 1}} = \dfrac{\sqrt{m^2 + 1}}{|m|} = \sqrt{5}$, so $1 + \dfrac{1}{m^2} = 5$, $m^2 = \dfrac{1}{4}$, $m = \pm\dfrac{1}{2}$.

The common tangents are $y = \dfrac{x}{2} + 2$ and $y = -\dfrac{x}{2} - 2$, i.e. $x - 2y + 4 = 0$ and $x + 2y + 4 = 0$. Putting $y = 0$ in each gives $x = -4$, so $T = (-4, 0)$.

The radius of C to the point of contact is along the normal of C, so KB_i is perpendicular to the tangent and B_i is the foot of the perpendicular from K. For $x - 2y + 4 = 0$: $B_1 = (1, 0) - \dfrac{1 + 4}{5}(1, -2) = (0, 2)$; similarly $B_2 = (0, -2)$.

$TK = 5$, and in the right triangle TB_1K: $TB_1 = \sqrt{TK^2 - r^2} = \sqrt{25 - 5} = 2\sqrt{5}$ (check: distance from $(-4, 0)$ to $(0, 2)$ is $\sqrt{16 + 4} = 2\sqrt{5}$).

The quadrilateral TB_1KB_2 is made of the two congruent right triangles TB_1K and TB_2K, so its

area is $2 \cdot \frac{1}{2} \cdot \sqrt{5} \cdot 2\sqrt{5} = 10$.

Final Answer: 10

Question 05

English

Let P_1 be the parabola $y^2 = 4(x+1)$. Let P_2 be the parabola that has the same focus and the same axis as P_1 and opens in the opposite direction. If the length of the common chord of P_1 and P_2 is 6, then the distance between the vertices of P_1 and P_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P_1 परवलय $y^2 = 4(x+1)$ है। माना कि P_2 वह परवलय है जिसकी नाभि तथा अक्ष P_1 के समान हैं तथा जो विपरीत दिशा में खुलता है। यदि P_1 तथा P_2 की उभयनिष्ठ जीवा की लम्बाई 6 है, तो P_1 तथा P_2 के शीर्षों के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$y^2 = 4(x+1)$ is $Y^2 = 4X$ with $X = x+1$, so $a = 1$, the vertex is $(-1, 0)$, the focus is $(-1+1, 0) = (0, 0)$ and the axis is the x-axis.

P_2 has focus $(0, 0)$, axis the x-axis, and opens towards the negative x-direction, so its vertex is at $(b, 0)$ with $b > 0$ (the focus lies inside, on the side towards which it opens), and its equation is $y^2 = -4b(x-b)$.

Points common to both satisfy $4(x+1) = -4b(x-b)$, i.e. $x + 1 = -bx + b^2$, $(1+b)x = b^2 - 1$, so $x = b - 1$. Hence the common chord is the line $x = b - 1$.

On this line $y^2 = 4(b - 1 + 1) = 4b$, so the end points are $(b-1, \pm 2\sqrt{b})$ and the length is $4\sqrt{b}$.

$4\sqrt{b} = 6$ gives $\sqrt{b} = \frac{3}{2}$, $b = \frac{9}{4}$.

The vertices are $(-1, 0)$ and $\left(\frac{9}{4}, 0\right)$, so the distance between them is $\frac{9}{4} + 1 = \frac{13}{4} = 3.25$.

Final Answer: 3.25

Question 06

English

Consider the ellipse $E : \frac{x^2}{4} + y^2 = 1$. Let C_1 and C_2 be the circles with centres $\left(-\frac{3}{10}, 0\right)$ and $\left(\frac{6}{5}, 0\right)$ respectively, each of which touches E at two distinct points. Then the length of the common chord of C_1 and C_2 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

दीर्घवृत्त $E : \frac{x^2}{4} + y^2 = 1$ पर विचार कीजिए। माना कि C_1 तथा C_2 क्रमशः केन्द्रों $\left(-\frac{3}{10}, 0\right)$ तथा $\left(\frac{6}{5}, 0\right)$ वाले वे वृत्त हैं जिनमें से प्रत्येक E को दो भिन्न बिन्दुओं पर स्पर्श करता है। तब C_1 तथा C_2 की उभयनिष्ठ जीवा की लम्बाई _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If a circle with centre $K(h, 0)$ touches E at Q, then the circle and E have the same tangent at Q, so they also have the same normal there; the normal of the circle at Q is the line QK. Hence K lies on the normal of E at Q.

For E: $a = 2$, $b = 1$, $e^2 = 1 - \frac{1}{4} = \frac{3}{4}$. The normal at $Q(2\cos\theta, \sin\theta)$ is $\frac{2x}{\cos\theta} - \frac{y}{\sin\theta} = 4 - 1 = 3$ (for $\sin\theta \neq 0$). It meets the x-axis at $x = \frac{3}{2}\cos\theta = \frac{3}{4}x_0$, where $x_0 = 2\cos\theta$.

So a circle centred at $(h, 0)$ touching E at points off the axis touches it at $(x_0, \pm y_0)$ with $x_0 = \frac{4}{3}h$ (two distinct points, which needs $|x_0| < 2$). A circle touching E only at a vertex $(\pm 2, 0)$ touches it at one point only, so the circles in the question are these.

C_1: $h = -\frac{3}{10}$, $x_0 = -\frac{2}{5}$, $y_0^2 = 1 - \frac{x_0^2}{4} = 1 - \frac{1}{25} = \frac{24}{25}$. Radius: $r_1^2 = (x_0 - h)^2 + y_0^2 = \frac{1}{100} + \frac{96}{100} = \frac{97}{100}$.

C_2: $h = \frac{6}{5}$, $x_0 = \frac{8}{5}$, $y_0^2 = 1 - \frac{16}{25} = \frac{9}{25}$. Radius: $r_2^2 = \left(\frac{8}{5} - \frac{6}{5}\right)^2 + \frac{9}{25} = \frac{4}{25} + \frac{9}{25} = \frac{13}{25}$.

Check of the double contact: $C_2 - E$ in the form $\left(x - \frac{6}{5}\right)^2 + y^2 - \frac{13}{25} - \left(\frac{x^2}{4} + y^2 - 1\right) = \frac{3}{4}\left(x - \frac{8}{5}\right)^2$, a perfect square, so the common chord of C_2 and E is the doubled line $x = \frac{8}{5}$; similarly $C_1 - E = \frac{3}{4}\left(x + \frac{2}{5}\right)^2$.

The circles meet, since the distance between centres $\frac{3}{2}$ lies between $r_1 - r_2$ and $r_1 + r_2 \approx 1.71$.

Common chord of C_1 and C_2: subtracting the equations, $\left(x + \frac{3}{10}\right)^2 - \left(x - \frac{6}{5}\right)^2 = \frac{97}{100} - \frac{52}{100}$, i.e. $\frac{3}{2}\left(2x - \frac{9}{10}\right) = \frac{9}{20}$, so $3x - \frac{27}{20} = \frac{9}{20}$ and $x = \frac{3}{5}$.

The distance of this line from the centre of C_2 is $\frac{6}{5} - \frac{3}{5} = \frac{3}{5}$, so the length of the common chord is $2\sqrt{r_2^2 - \frac{9}{25}} = 2\sqrt{\frac{4}{25}} = \frac{4}{5} = 0.8$. (End points $\left(\frac{3}{5}, \pm\frac{2}{5}\right)$.)

Final Answer: 0.80